AdaBoost.R2 Algorithm (Regression Boosting)

Input:

- Training data: $\{(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)\}$
- Weak learner (e.g., decision tree regressor having depth=1 (stump))
- Number of estimators/weak-learners M

Step 1: Initialize weights

Assign equal weights to all samples:

$$w_i^{(1)} = \frac{1}{N}, \quad i = 1, 2, \ldots, N$$

Step 2: For each boosting round $m = 1, 2, \ldots, M$:

1. Train weak learner $h_m(x)$ on the weighted dataset.
2. Compute absolute errors:

   $$e_i = |y_i - h_m(x_i)|$$

   Normalize errors to $[0, 1]$:

   $$L(y_i, h_m(x_i)) = \frac{|y_i - h_m(x_i)|}{\max_j |y_j - h_m(x_j)|}$$

3. Compute weighted error:

   $$e_m = \frac{\sum_{i=1}^{N} w_i^{(m)} L(y_i, h_m(x_i))}{\sum_{i=1}^{N} w_i^{(m)}}$$

4. Compute model weight:

   $$\beta_m = \frac{e_m}{1 - e_m}$$

   (valid if $e_m < 0.5$; otherwise stop boosting)

5. Update sample weights:

   $$w_i^{(m+1)} = w_i^{(m)} \cdot \beta_m^{(1 - L(y_i, h_m(x_i)))}$$

6. Normalize:

   $$w_i^{(m+1)} \leftarrow \frac{w_i^{(m+1)}}{\sum_{j=1}^{N} w_j^{(m+1)}}$$

Step 3: Final Prediction

Each weak learner produces a prediction $h_m(x)$. combine the predictions using weighted mean:

$$H(x) = \frac{\sum_{m=1}^{M} \alpha_m \, h_m(x)}{\sum_{m=1}^{M} \alpha_m}$$

where $\alpha_m = \log \frac{1}{\beta_m}$